Extensions of residues containing subfield groups and tori

Human Compression project

Abstract

We determine all rank-four residually weakly primitive extensions of the specified residues R10, R11 and R13 in the fixed higher-rank Suzuki paper. A torus subgroup together with its full Weyl complement determines its containing subfield groups uniquely. A second argument extracts a Boolean divisor lattice from residual weak primitivity. Neither argument assumes the source's classification of all residues.

1 Premises and residual conditions

Let $S_f = \mathrm{Sz}(2^f)$, with f odd, denote a standard subfield group of S_N, where $f \mid N$. We use the classical subgroup and conjugacy theorems for Suzuki groups, stated in the accepted preliminary papers. In particular, every proper nonsoluble subgroup is a simple subfield group. Simple subfield groups are self-normalizing, and their copies inside any given Suzuki group form one conjugacy class. The other maximal subgroups are point stabilizers and normalizers of the three cyclic torus types. A nonidentity odd-order element determines its torus uniquely. The three torus orders

$$2^f - 1, \qquad 2^f + 2^{(f+1)/2} + 1, \qquad 2^f - 2^{(f+1)/2} + 1$$

are pairwise coprime. Their normalizers have respective complements of orders $2, 4, 4$. In each case the complement contains an involution inverting the torus. Copies of the standard subfield groups intersect according to $S_a \cap S_b = S_{\gcd(a,b)}$.

The accepted torus construction supplies two compatible nonsplit families T_f^+, T_f^-, together with a common complement of order four. Their signs need not agree with the signs of the square-root formulas at every exponent. What matters here is

$$T_a^\epsilon \cap T_b^\epsilon = T_{\gcd(a,b)}^\epsilon, \qquad \gcd(|T_a^\epsilon|, |T_b^\epsilon|) = |T_{\gcd(a,b)}^\epsilon|. \tag{1}$$

For $d \mid f$, the two bottom orders divide the two different upper orders. These are reused structural premises, proved in the accepted paper *Subfield and torus geometries*.

Write $G_J = \bigcap_{j \in J} G_j$ and $G_\varnothing = G$. Residual weak primitivity means that for each proper $J \subset I$ some $G_{J \cup \{i\}}$ is maximal proper in G_J. It implies

$$G_{J \cup \{i\}} < G_J \quad (i \notin J). \tag{2}$$

Indeed, equality would imply $G_{I \setminus \{i\}} \leq G_i$; the last panel would then equal G_I, contrary to maximal proper inclusion. The arguments below only need this group condition.

2 Uniqueness of subfield overgroups

Lemma 1. *A fixed simple subfield group is contained in at most one subfield group of any prescribed degree. The same assertion holds for $H = C_m : C_c$ when $m > 1$ is odd and C_c is the full torus complement: $c = 2$ in a split torus, or $c = 4$ in a nonsplit torus.*

Proof. The first assertion follows from conjugacy of subfield copies and self-normalization. More generally, suppose $H \leq S_f$ and a conjugate gS_fg^{-1} also contains H. If the copies of H inside S_f are conjugate, adjust g by an element of S_f to normalize H. When $N_G(H) \leq S_f$, this forces $gS_fg^{-1} = S_f$.

For the stated torus subgroup, C_m is characteristic: it consists of the identity and all odd-order elements of H. Therefore its normalizer is contained in the normalizer $T : C_c$ of its unique torus. Remove the complement component of a normalizing element, leaving $h \in T$. If $z \in C_c$ is the involution, then $hzh^{-1} = h^2z$ under a choice of conjugation convention. Normalization of H implies $h^2 \in C_m$. Since $|T|$ is odd, this implies $h \in C_m$. Hence $N_G(H) = H$.

Copies of H in a Suzuki group are conjugate: the odd subgroup selects a unique torus type, the cyclic torus has a unique subgroup of order m, and its complements are conjugate in the odd-order torus extension. This proves the second assertion by the preceding transporter argument. The use of the full complement is essential; the assertion does not cover $C_m : C_2$ inside a nonsplit torus. □

Lemma 2. *Suppose two simple subfield groups of degrees a, b contain the same $H = C_m : C_c$ of Lemma 1. In the nonsplit case assume $m > 5$. Their intersection is a simple subfield group of degree $d = \gcd(a, b)$.*

Proof. The split type persists in every odd-degree subfield containing it: a nonsplit occurrence would make its odd order divide both $2^N - 1$ and $2^{2N} + 1$, whose gcd is one. In the split case $m \mid \gcd(2^a - 1, 2^b - 1) = 2^d - 1$. Thus H has a standard copy in S_d. Here $d > 1$, since $m > 1$. In the nonsplit case H is nonsplit in both groups, since its complement has order four. The identity

$$\gcd(2^{2a} + 1, 2^{2b} + 1) = 2^{2d} + 1$$

holds because a/d and b/d are odd. Thus m divides the product of the two nonsplit orders at d. It divides just one of them: each bottom order divides a different upper order at a, whereas m divides only the upper order containing its torus. Coprimality of the upper orders excludes any factor from the other bottom order. The compatible torus family therefore places a copy of H in S_d. Its action agrees with that of H: the order-four action is raising to $\pm 2^d$, and an odd multiple of d gives this action or its inverse modulo m. Reversing a complement generator identifies the two presentations. Since the only nontrivial nonsplit order at $d = 1$ is 5, the assumption $m > 5$ makes $d > 1$.

Conjugate the copy of H in $S_d \leq S_a$ onto the given one inside the first group. In these coordinates the standard S_b is an overgroup of H. Lemma 1 identifies it with the second group. Their intersection is now the standard S_d. □

Lemma 3. *Let $G_0 = S_{N/c}$ be maximal in S_N, where c is prime. Suppose a proper simple subfield group K shares an embedded simple subfield with G_0, and $G_0 \cap K = S_{N/(ca)}$, where a is prime and the common group is simple. If $K \not\leq G_0$, then $a \neq c$ and $K = S_{N/a}$ after simultaneous standardization.*

Proof. The common simple group and Lemma 1 make both groups standard. If v is the degree of K, then $v \mid N$, $v \nmid N/c$, and $\gcd(v, N/c) = N/(ca)$. Since N/c lacks just one occurrence of the prime c, the first two conditions force $v_c(v) = v_c(N)$. Consequently $a \neq c$, as otherwise the gcd would have to lack a second occurrence of c. At a the gcd equality forces $v_a(v) = v_a(N) - 1$; at every other prime different from c it forces the full valuation in N. Thus $v = N/a$. This proof allows a and c to divide the common base exponent. □

3 A divisor lattice forced by residual primitivity

Lemma 4. *Let $L = C_M : C_c$ have odd rotation order M. Let $H_i = C_{d_i} : C_c$ for $1 \le i \le r$, with a common fixed complement, and put $\gamma = \gcd(d_1, \dots, d_r)$. If these intersections satisfy residual weak primitivity, then there are distinct primes $\ell_1, \dots, \ell_r$ such that*

$$M = \gamma \prod_{i=1}^{r} \ell_i, \qquad d_i = M/\ell_i. \tag{3}$$

Primes ℓ_i may divide γ.

Proof. For $r = 1$ the conclusion is precisely maximality; assume $r \ge 2$. Every subgroup between $C_d : C_c$ and $C_M : C_c$ has the form $C_h : C_c$, $d \mid h \mid M$: it contains the complement, and removing that component from each element leaves its rotation. Thus a proper inclusion is maximal exactly when the ratio of rotation orders is prime. Successively applying residual weak primitivity gives r prime-index steps from L to the full meet. Hence $\Omega(M/\gamma) = r$, with multiplicity counted.

For each i, (2) says $\gcd(d_j : j \ne i) > \gamma$. Some prime ℓ_i therefore has

$$v_{\ell_i}(d_i) < \min_{j \ne i} v_{\ell_i}(d_j).$$

One prime cannot witness this inequality for two different indices. We have found r distinct primes dividing M/γ. Since its total number of prime factors is r, each occurs exactly once and there are no other prime factors. At the witnessing prime, d_i has the bottom valuation and every other d_j the top valuation. At all other primes every d_j has the valuation of M. These statements give (3), including when a witnessing prime divides γ. □

4 The torus residues R10 and R11

The R11 row calls n, n' two primes without expressly distinguishing them. The prescribed lattice forces them to be distinct: if $n = n'$, the subgroups G_{012}, G_{023} have the same cyclic rotation subgroup inside G_{02} and contain the same full complement G_{0123}. They coincide, so their meet has order $c\gamma n$ instead of the prescribed $c\gamma$. Thus the equal-prime case has no realization.

Theorem 5. *Let $u = 2^e \ge 8$ and let a, b be distinct odd primes. Suppose a rank-four residually weakly primitive pregeometry in $\mathrm{Sz}(q)$ has a maximal parabolic $G_0 = \mathrm{Sz}(u^{ab})$ with the prescribed residue*

$$G_{01} = \mathrm{Sz}(u^a), \quad G_{03} = \mathrm{Sz}(u^b), \quad G_{02} = C_{\gamma nn'} : C_c,$$
$$G_{013} = \mathrm{Sz}(u), \quad G_{012} = C_{\gamma n} : C_c, \quad G_{023} = C_{\gamma n'} : C_c, \quad G_{0123} = C_\gamma : C_c, \tag{4}$$

where n, n' are distinct primes. For R10 put $c = 2$, $\gamma = u - 1$; for R11 put $c = 4$ and let γ be a nonsplit torus order at u. Then an odd prime d distinct from a, b satisfies

$$q = u^{abd}, \qquad G_1 = \mathrm{Sz}(u^{ad}), \qquad G_3 = \mathrm{Sz}(u^{bd}), \qquad G_{13} = \mathrm{Sz}(u^d).$$

There is a prime n'' distinct from n, n' such that

$$\begin{aligned} G_2 &= C_{\gamma nn'n''} : C_c, \\ G_{12} &= C_{\gamma nn''} : C_c, \qquad G_{23} = C_{\gamma n'n''} : C_c, \\ G_{123} &= C_{\gamma n''} : C_c. \end{aligned} \tag{5}$$

These formulas hold simultaneously after conjugation. In R10, $n'' \mid (u^d - 1)/(u - 1)$. In R11, $\gamma n''$ divides the compatible nonsplit torus order at u^d, and $\gamma nn'n''$ divides its ambient order. All remaining intersections are determined by (4)–(5).

Proof. The groups G_1, G_3 contain the simple groups G_{01}, G_{03}, so are simple subfield groups. All share G_{013}, allowing simultaneous standardization by Lemma 1. Maximality of G_0 gives $q = (u^{ab})^d$ for a prime d. Applying Lemma 3 to its two specified maximal subfield intersections gives $d \neq a, b$, and the asserted degrees and meet. Conjugate the common full torus normalizer G_{0123} inside S_e onto the chosen standard normalizer. This preserves all three standard simple groups and fixes their torus coordinates simultaneously.

Put $H = G_{02}$. If G_2 were a simple subfield group, Lemma 2 would make $G_0 \cap G_2$ a simple subfield group, contradicting $G_0 \cap G_2 = H$. In the nonsplit case its rotation order $\gamma nn'$ exceeds five even when $\gamma = 5$. This explains why one must use the entire H, rather than claim that its base rotation C_γ generates the base field.

Choose a minimal Suzuki overgroup of G_2, allowing the ambient group. Since G_2 is not itself a Suzuki group, it lies in a maximal subgroup of this overgroup. Minimality rules out the subfield type, so it lies in a point stabilizer or torus normalizer. A point stabilizer cannot contain H: its quotient by its normal two-subgroup is cyclic of odd order, whereas the involution in H inverts a nontrivial odd rotation. The images of that involution and its inverse action would force this rotation to equal its inverse. It cannot disappear in the two-subgroup either. The wrong torus types are excluded by coprimality. The unique torus containing the odd part of H is therefore also the torus for G_2. Thus $G_2 = C_M : C_c$, with the same complement as H.

Its three parabolic intersections with G_0, G_1, G_3 all contain the complement from G_{0123}. Their full meet is $C_\gamma : C_c$. Lemma 4 says that M/γ is the product of three distinct primes, one omitted at each parabolic. The prescribed orders of G_{02}, G_{012}, G_{023} identify two as n, n' and force the third, n'', in precisely the positions of (5). Intersecting with $G_{13} = \mathrm{Sz}(u^d)$ proves the final divisibility condition. The inherited unique torus determines its compatible branch. Its containment in the ambient torus also gives the asserted ambient divisibility. □

5 The pure subfield residue R13

Theorem 6. *Suppose $G_0 = S_{N/d}$ is maximal in S_N and its prescribed rank-three residue has parabolics $G_{0i} = S_{N/(da_i)}$ for $i = 1, 2, 3$, where a_1, a_2, a_3 are distinct odd primes and their common intersection is simple. Every rank-four residually weakly primitive extension has $a_i \neq d$ and $G_i = S_{N/a_i}$, after one simultaneous conjugation. All intersections are the standard subfield intersections.*

Proof. Each G_i contains the simple G_{0i}, hence is a simple subfield group. The common simple subgroup standardizes every one of them by Lemma 1. Strictness excludes $G_i \leq G_0$. Lemma 3 now gives the stated degrees. Their simultaneous standard coordinates give every intersection by the gcd formula. □

6 Source scope and attribution

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20, fixed edition, pp. 14–15. The R10 paragraph interchanges the labels of its second new Suzuki parabolic and its dihedral parabolic. The consistent labels are G_1, G_3 for the Suzuki groups and G_2 for the dihedral group, as fixed by the prescribed intersections in its opening sentence.

The ambient structure and torus compatibility are borrowed as stated. The overgroup, divisor-lattice and conditional extension arguments above are supplied here to repair a gap in our first claim ledger. A refutation of the source's global classification did not settle these distinct conditional assertions. The present theorems settle the rank-four assertions for

the actual prescribed residues. The accepted constructions already prove realization of their admissible canonical parameters. Neither those existence results nor this note classifies all unrestricted higher-rank relative positions. In particular, the source's separate upper rank bound for pure subfield constructions remains refuted by the previously accepted repeated-prime example. No new open-conjecture, principal-refutation, Lean or compression credit is asserted here.